

Supply Chain Disruption Assistant & Fleet Utilisation Optimizer

AI-powered **logistics monitoring** and decision support

Team **CodeVault** | **AI Track**

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The Problem

Logistics teams face fragmented data and **disconnected processes**.

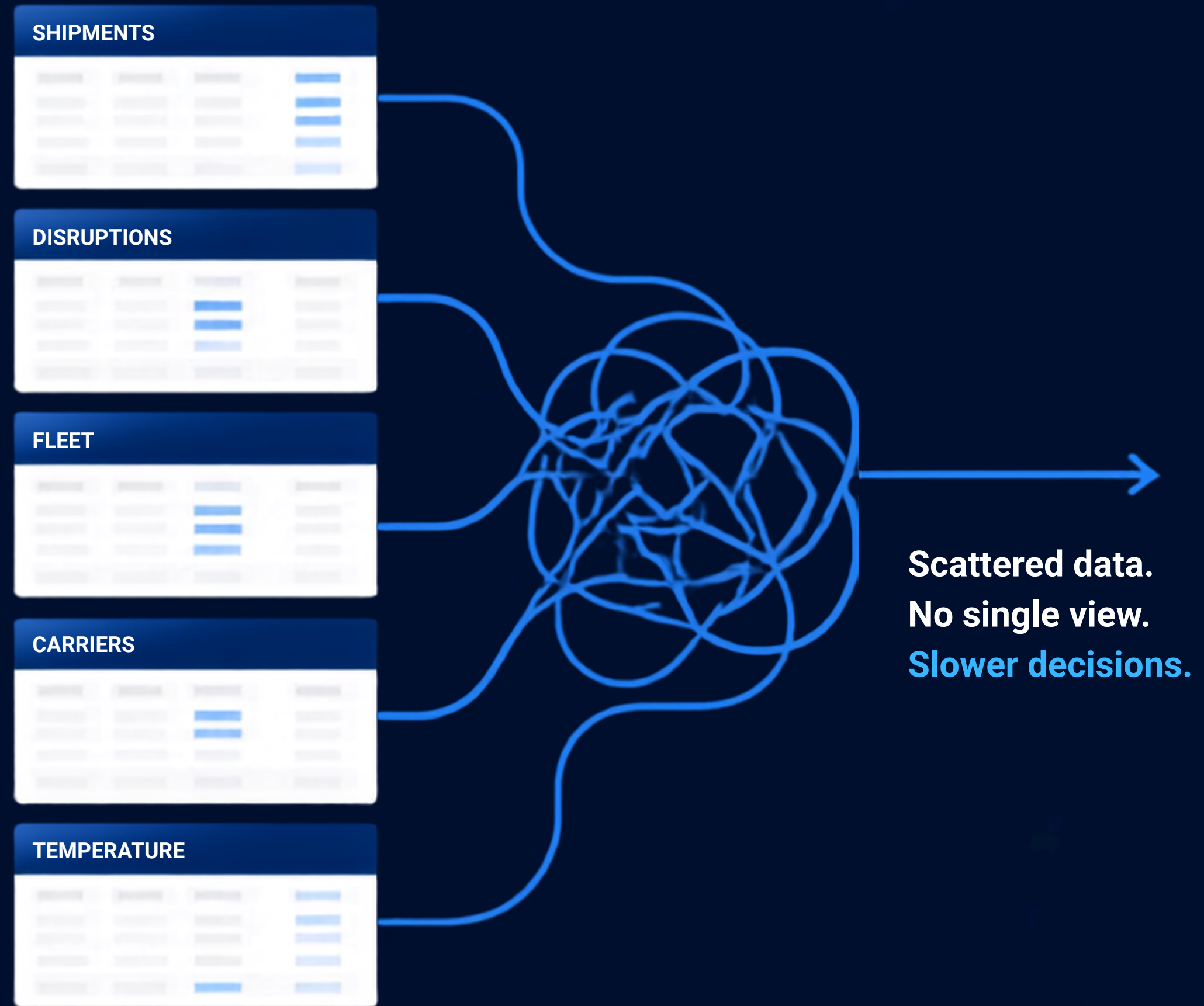
Identifying shipments affected by disruptions

Detecting underutilized fleet vehicles

Monitoring cold-chain temperature risks

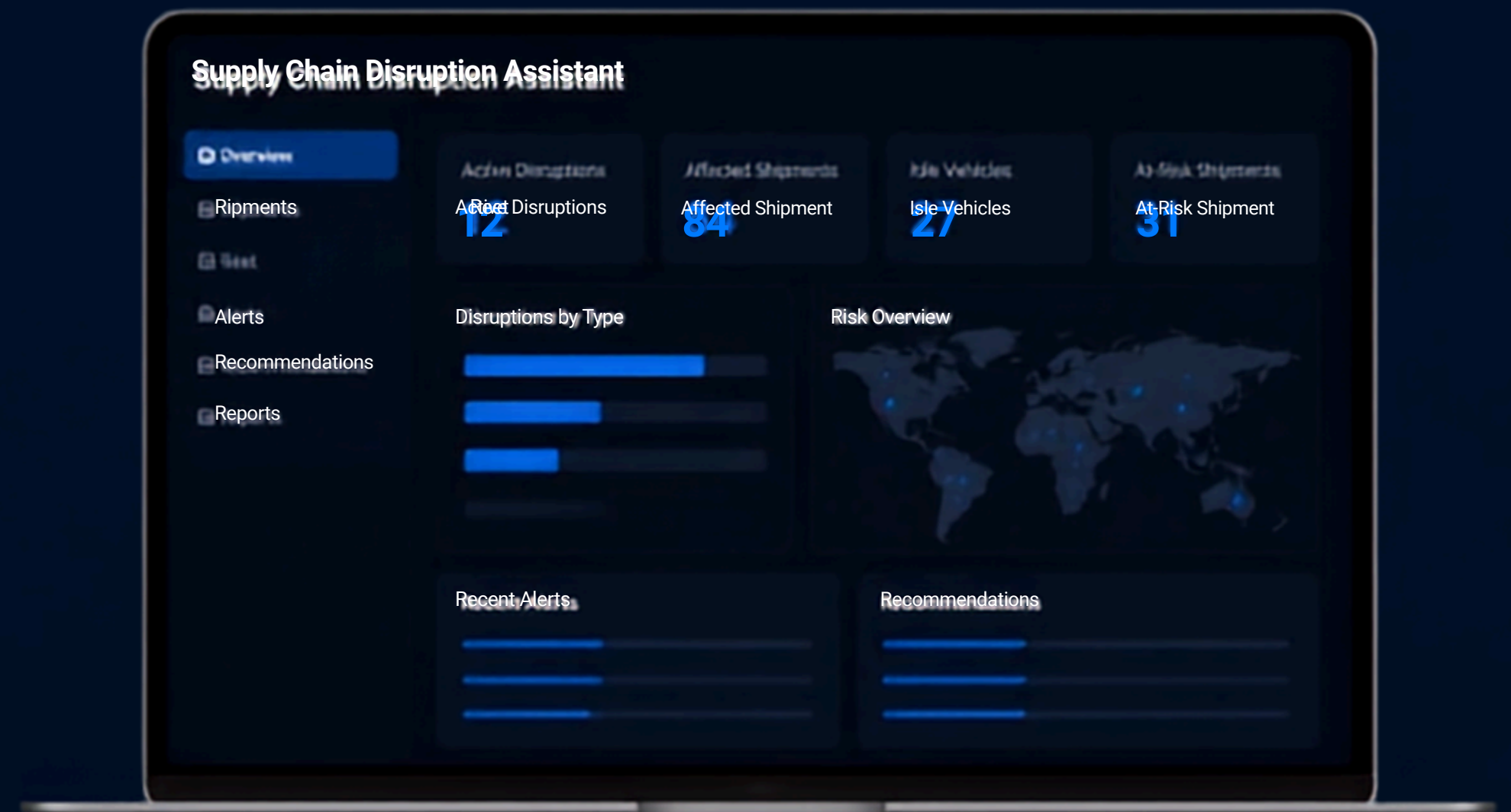
Finding alternative carriers and rerouting options

Operational data scattered across multiple records **hampers fast decision-making**.



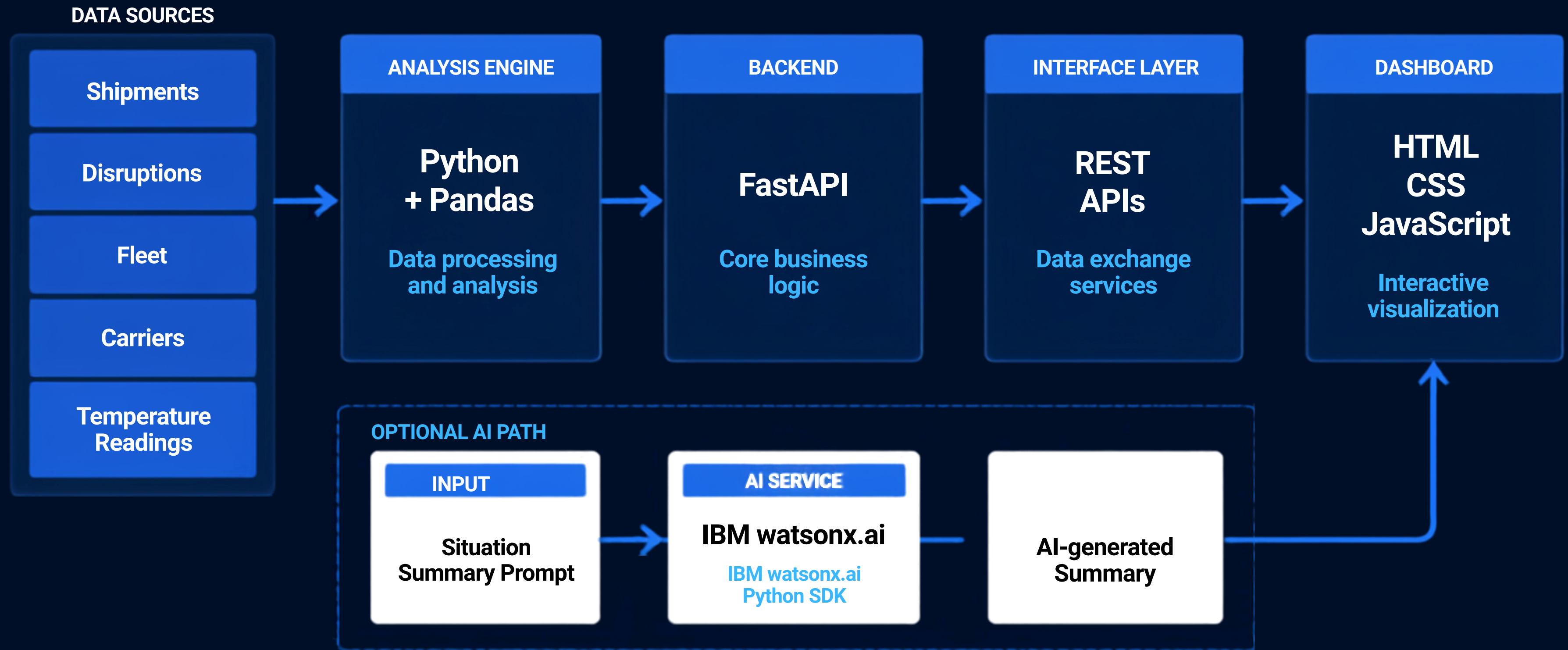
Our Solution

An AI-powered assistant built on FastAPI that turns operational data into **actionable insights** via a web dashboard.



System Architecture

Data flows from multiple sources through our analysis engine to deliver insights and **optimized decisions**.



Live Dashboard & Key Features

Real-time visibility into disruptions, performance, and recommended actions.

4

Affected Shipments

4

Idle Fleet Vehicles

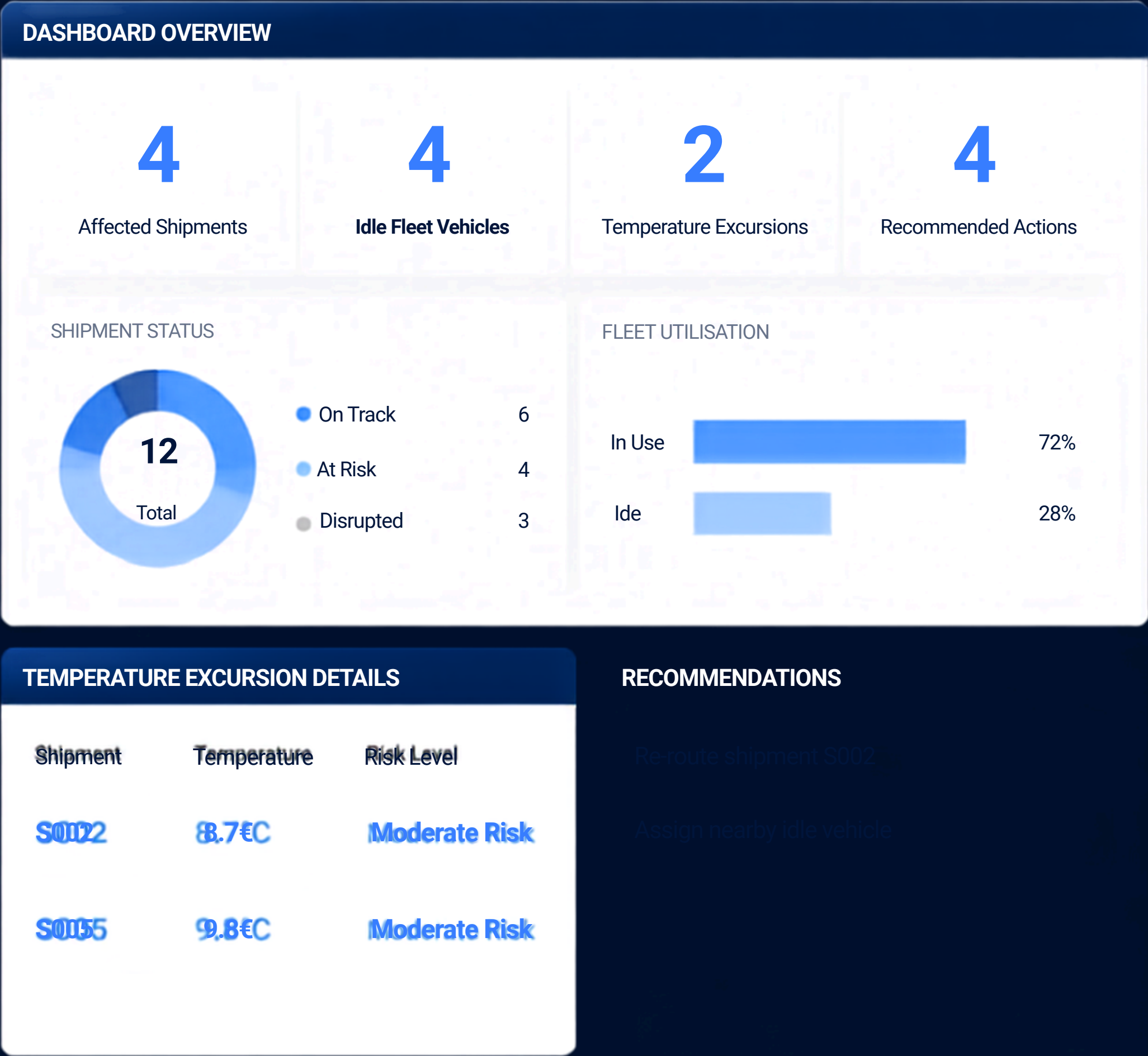
2

Temperature Excursions

4

Recommended Actions

One view to spot issues fast and act with confidence.



IBM Technologies

IBM | watsonx.ai

Integrated via FastAPI backend with
IBM watsonx.ai Python SDK

Generates concise **natural-language**
operational situation summaries

Uses **“ibm/granite-3-8b-instruct”** model
flet credentials configured

Core detection logic remains **deterministic**

Graceful fallback when credentials
unavailable; dashboard remains **fully**
functional

IBM | Bob

AI coding assistant used
during development

Helped with **code inspection** and
automated test suite creation/expansion

Tests cover **disruption detection, fleet**
utilization, temperature excursions,
recommendations, data integrity

Current test suite: **45 passing**
automated tests



Results & Impact

KEY METRICS

45

Tests Passed

4

Affected Shipments

4

Idle Fleet Vehicles

2

Temperature Excursions

4

Recommendations

IMPACT SUMMARY

Prototype consolidates raw logistics data into a unified operational view enabling quick risk identification and decision support.

FUTURE IMPROVEMENTS

Live GPS and IoT data integration
Real-time temperature monitoring
Predictive disruption detection
Traffic and route intelligence incorporation
Production-grade database and cloud deployment
Expanded AI integration

Team CodeVault

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Turning logistics data into **faster,**
smarter operational decisions.

